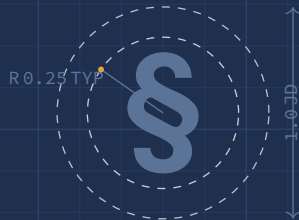


FIELD GUIDE // PART 01 OF 03

TEXT AS DATA

LEGAL TEXT AS VECTORS



02

The question, restated

- Which of the 2,674 other judgments is most similar to NJA 2019 s. 684/B 1064-19, the **anchor case**?
- This part: two *word-based* readings of similarity.
- Both start from the same fact — an opinion is a sequence of words with a meaning — and both reduce that to a number.
- the number is only useful if we know what it assumes

judgment text215,550 paragraphs
one document per judgment2,676 docs
Swedish, pre-coded into sections
(facts, reasoning, verdict, ...)
similarity will be: a number
between 0 and 1

The idea: bag of words

- First step: **tokenise** — split the text into words.
- **Stem** — collapse inflections to a root, so that *domar* and *domen* count as one stem.
- **Drop stopwords** — but keep legal vocabulary on purpose.
- The judgment becomes a *multiset* of stems: word order and structure are thrown away.

```
domar .....dom
domen .....dom
domstol .....domstol
yrkanden .....yrk
överklagandet .....överklag
```

Snowball Swedish stemmer: one stem
per word family

```
the buyer did not breach
the buyer breached
stem + drop stopwords (did, not)
{buyer, breach} .....{buyer, breach}
same bag .....negation lost
```

TF-IDF: weigh words by distinctiveness

- **tf** — how often a stem occurs in this judgment.
- **idf** — how rare the stem is across the corpus:
 $idf = \log(N/\text{docs with stem})$.
- **tf-idf** = $tf \times idf$: a word matters when it is frequent here *and* unusual overall.
- A judgment is then a **vector** of its stem weights — one dimension per distinct stem.

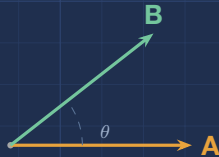
2,676 documents
114,660 distinct stems
(>100,000 dimensions!)

"mopedbil"rarehigh idf
"rätt"everywherelow idf

most cells are zero --- the matrix
is stored sparse

Cosine: how much two vectors point the same way

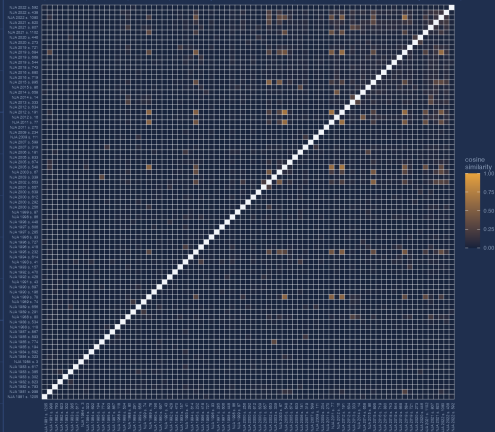
- Two judgments are similar if their vectors point in the same *direction*.
- **Cosine** = angle between the vectors: 1 identical direction, 0 orthogonal, -1 opposite.
- Length-invariant: a long judgment can be compared to a short one.
- Doing this for every pair gives the full $2,676 \times 2,676$ similarity table — *in seconds*.



$$\text{COSINE}(A, B) = 1 \text{ WHEN } \theta = 0$$

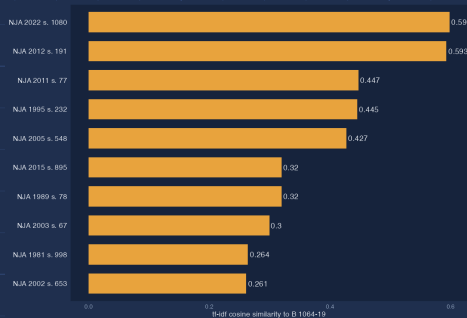
The full table — a slice

- Every cell: cosine similarity between two judgments, computed from their tf-idf vectors.
- Bright = similar; dark = unrelated.
- The anchor's row (bottom edge) is what the demo reads out.
- no judgement yet --- just geometry on word counts



What tf-idf returns

- A **ranked list**: every other judgment scored against the anchor, best first.
- Top hit: B 5404-22, NJA 2022 s. 1080 — drunk driving with an A-traktor; same area, same question.
- The output is a number and a list — *no argument*.
- read off the list ... and then ask: does a high score mean legally similar?



Blind spots of exact-word matching

- Word counts see *words*, not *who did what to whom*.
- Four statements, two lawful meanings:
 - s_1 The **seller breached** the contract.
 - s_2 The **buyer breached** the contract.
 - s_3 The vendor failed to perform the agreement.
 - s_4 The seller **had not breached** the contract.
- tf-idf cosine: $s_1 \approx s_2 \approx s_4$ (same words), while s_3 (same meaning) scores zero.
- surface similarity $\neq$ legal similarity

pair	cosine	what it means
s_1, s_2	≈ 1	role swap, missed
s_1, s_3	≈ 0	paraphrase, missed
s_1, s_4	≈ 1	negation, missed

THE SAME LIMIT APPLIES TO THE REAL
JUDGMENTS

Doc2vec: a space where meaning pulls together

- **Word2vec** idea: words used in similar contexts get similar vectors — *säljare* and *säljer* neighbours.
- **Doc2vec** adds one vector *per document*, learned jointly while predicting the words (PV-DBOW).
- Here: trained on the *facts* sections, 50 dimensions, a few minutes of compute.
- Judgments can now be similar without sharing a single word.

anchor, nearest neighbours:

B 1099-21	0.880
B 5404-22	0.785
B 6092-20	0.693
B 1123-87	0.692
B 4564-21	0.689

2,246 × 50-dimensional vectors from
the facts of each case

Two word-based readings, two costs

TF-IDF + cosine

- transparent — every weight is inspectable
- cheap — no training, runs in seconds
- language-agnostic
- **exact words only**: no synonyms, no order, no negation

Doc2vec + cosine

- synonym-aware — paraphrase gets partial credit
- no hand-made features
- **opaque** — the numbers are learned, not inspectable
- **training data & settings matter**: 50 dims vs 300, facts vs everything

- Overlap between the two top-10 lists: exactly **1 of 10** (B 5404-22, the A-traktor case).
- Both readings answer *words similar* — not *legally similar*. For that we need other evidence.

Live demo: text as data in R

- Loads `text.csv` + `cases.csv` — no database, no setup.
- Corpus -> tokens -> stems -> tf-idf matrix -> cosine.
- Then `doc2vec` on the facts: `paragraph2vec`, 50 dims.
- Both top-10 lists are saved for the comparison in part 04.
- the same script is in your kit --- run it after the break

```
$ Rscript 01_text_analysis.R  
[1] corpus 2676 judgments, 215550  
paragraphs [2] tf-idf 2676 docs ×  
114660 stems cosine vs anchor: B  
5404-22 0.598 [3] doc2vec (facts,  
50 dims) 2246 vectors, top: B  
1099-21 0.880 saved: top_10_tfidf,  
top_10_doc2vec .....
```